A Four-Position Partition of Morphisms in Elementary Topoi with Distinction Structure

Chris Brink github.com/thefalsework/papers **Version.** v1.2 (repository text), October 2026; v1.1, September 2026 (first archived deposit) — preprint, not yet submitted to a venue. v1.0 was the May 2026 draft; v1.1 changes no mathematical content and adds the deposit metadata, the citation block, and the companion-results postscript recording lattice-level developments (the non-degeneracy criterion, the perceptron bridge) that postdate the May draft. v1.2 (October 2026, repository text; not yet deposited) adds Remark 5.3′: the partition is lopsided because triple negation collapses, so the outside of a distinction is always regular and only the inside can carry a closure-residue. No other change.

Abstract

We define a *distinction structure* on a category as an idempotent endofunctor with unit, with monadic idempotency as a categorical rendering of the calling axiom of Spencer-Brown's *Laws of Form* (Spencer-Brown 1969). In an elementary topos equipped with a non-trivial distinction structure, we prove that the morphism space partitions into exactly four pairwise-disjoint structural classes, characterized by Heyting conditions on the morphism's image relative to the kernel image (the image of the unit). The four classes are mutually exclusive and jointly exhaustive over morphisms with non-trivial image. The proof reduces to standard Heyting algebra identities and a case analysis on lattice position. The fourth class — the *closure-residue* between a subobject and its double-negation — exists specifically as a feature of non-Boolean intuitionistic structure and collapses in Boolean settings. The construction is formalized in Lean 4 against Mathlib and kernel-checked; the underlying Heyting algebra instance on subobjects of an elementary topos was submitted to Mathlib as PR #39618 (closed unmerged under the project's new-contributor AI-use policy; the instance remains kernel-checked in this development).

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1. Introduction

The subobject lattice Sub(X) of an object X in an elementary topos carries a Heyting algebra structure (Mac Lane and Moerdijk 1992, IV.6). When the topos is non-Boolean, this Heyting algebra exhibits a specific asymmetry: for a subobject $a \in \mathrm{Sub}(X)$, the double-negation closure $\neg\neg a$ need not equal a, and the strict difference between them — the *closure-residue* — is in general non-trivial.

This paper identifies a structural partition of morphisms into an elementary topos that exploits this closure-residue. We work with an additional categorical structure — a *distinction structure* (D, η, ι) consisting of an endofunctor D with unit η and a coherent idempotency $D \ggg D \cong D$ — that encodes the calling axiom of Spencer-Brown (1969) in categorical form. The *kernel image* at an object Y is the subobject $\mathrm{Im}(\eta_Y) \in \mathrm{Sub}(D(Y))$, and we ask how the D-image of an arbitrary morphism relates to this kernel image.

We prove that, under the assumption that D is non-trivial (η is not a natural isomorphism), every morphism with non-trivial D-image satisfies exactly one of four pairwise-disjoint Heyting conditions:

- the image lies inside the kernel image (**Infrastructure**),
- the image straddles the kernel image and its complement (**Distribution**),
- the image lies in the closure-residue — inside the double-negation closure but not in the kernel image itself (**Exploitation**),
- the image lies in the Heyting complement (**Refusal**).

The proof is a case analysis whose disjointness reduces to the Heyting identities $a \sqcap a^c = \bot$ and $a^c \sqcap a^{cc} = \bot$, and whose exhaustiveness follows from the trichotomy of how a subobject can position itself relative to another in a Heyting algebra.

The fourth class (Exploitation) is structurally distinctive: it exists as a separate region of the lattice only because $\neg\neg a$ need not equal a. In a Boolean topos the closure-residue collapses, Exploitation merges with its neighbors, and the partition reduces to three classes. The four-fold structure is a specific feature of non-Boolean intuitionistic logic.

The construction is implemented in Lean 4 against Mathlib4 (Mathlib Community 2026) and verified by the kernel. The underlying Heyting algebra instance on Sub(X) for elementary topoi — required by the partition theorem and missing from Mathlib's standard development — was submitted upstream as PR #39618 (closed unmerged on 2026-05-28 under Mathlib's new-contributor AI-use policy; it remains kernel-checked in this development) and follows the Mac Lane–Moerdijk equalizer-residual construction (Mac Lane and Moerdijk 1992, IV.6 Proposition 2).

The paper is organized as follows. Section 2 establishes notation and recalls the relevant categorical structure. Section 3 defines the distinction structure and the kernel image. Section 4 states the four position predicates. Section 5 proves the partition theorem. Section 6 discusses the formalization. Section 7 addresses motivation, related work, and scope limits.

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2. Preliminaries

We work throughout in a category $\mathcal{C}$ satisfying the elementary-topos hypothesis bundle currently exposed in Mathlib:

- **HasClassifier**: $\mathcal{C}$ has a subobject classifier Ω with universal morphism true: $1 \to \Omega$.
- **HasPullbacks**: $\mathcal{C}$ has pullbacks of all cospans.
- **HasEqualizers**: $\mathcal{C}$ has equalizers of all parallel pairs.
- **HasInitial**: $\mathcal{C}$ has an initial object 0.
- **InitialMonoClass**: the unique morphism $0 \to X$ is monic for every X.
- **HasImages**: every morphism in $\mathcal{C}$ has an image factorization.
- **HasBinaryCoproducts**: $\mathcal{C}$ has binary coproducts.

This bundle is entailed by $\mathcal{C}$ being an elementary topos in the standard sense (Mac Lane and Moerdijk 1992, IV.1; Johnstone 2002a). We work with the explicit bundle rather than a unified elementary-topos typeclass because current Mathlib exposes the constituents rather than a single typeclass for the bundled signature.

For $X \in \mathrm{Ob}(\mathcal{C})$, $\mathrm{Sub}(X)$ denotes the lattice of subobjects of X. Under the stated hypotheses, $\mathrm{Sub}(X)$ carries a Heyting algebra structure (Mac Lane and Moerdijk 1992, IV.6 Proposition 2): meet is given by pullback, the bottom subobject is the image of the unique morphism from 0, the top subobject is X itself, the implication $P \Rightarrow Q$ is the residual constructed as the equalizer of the characteristic morphisms $\chi_{P \sqcap Q}$ and χ_P, and the complement is $\neg P = (P \Rightarrow \bot)$.

This Heyting structure is in general non-Boolean. The double-negation operation $P \mapsto \neg\neg P$ satisfies $P \le \neg\neg P$ always, with equality iff $\mathrm{Sub}(X)$ is Boolean at P. We write P^c for the Heyting complement $\neg P$ and P^{cc} for the double-negation closure $\neg\neg P$. The strict closure-residue at P is the structural region between P and P^{cc}; in non-Boolean settings, this region is non-trivial.

For $f: X \to Y$ in $\mathcal{C}$, we write $\mathrm{Im}(f)$ for the image of f as a subobject of Y, via the image factorization that HasImages provides.

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3. Distinction Structure

Definition 3.1. A *distinction structure* on a category C is a triple (D, η, ι) where:

- D: C ⇒ C is an endofunctor,
- η: 1_C ⟹ D is a natural transformation,
- ι: D ⋙ D ≅ D is a natural isomorphism

satisfying the coherence condition

η_{D(X)} ≫ ι.hom_X = 1_{D(X)}

for every X ∈ Ob(C).

The functor D is the *distinction operation*; η is the *unit* of the distinction; ι is the *idempotency*. The coherence condition relates the two natural morphisms from D(X) to D(X) and reflects the calling axiom of Spencer-Brown (1969): applying the distinction operation to an already-distinguished object is structurally equivalent to the original distinction. Monadic idempotency D ⋙ D ≅ D, together with the coherence condition, is one categorical rendering of Spencer-Brown's calling axiom (⟨ ⟩⟨ ⟩ = ⟨ ⟩, with the juxtaposition of marks translated to sequential functor composition). Other categorical readings of the calling axiom are possible (see Kauffman 1987 for an alternative recursive-form treatment); the framework's specific rendering is one option in that space, chosen here for its alignment with monad theory and reflective subcategories (Remark 5.5).

Definition 3.2. A distinction structure (D, η, ι) is *non-trivial* if there exists X ∈ Ob(C) such that η_X is not an isomorphism in C.

Non-triviality is the assumption that the distinction operation produces structural difference somewhere. If η were a natural isomorphism, the distinction would be vacuous and the partition theorem would degenerate.

Definition 3.3. For Y ∈ Ob(C), the *kernel image* at Y is the subobject

a_Y := Im(η_Y) ∈ Sub(D(Y)),

via the image factorization of the unit at Y.

The kernel image is the territory in D(Y) that the unit reaches when applied at Y. Sub(D(Y)) is the Heyting algebra in which the partition predicates will be stated.

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4. The Four Position Predicates

For a morphism f: X ⟶ Y in C, write img_f := Im(D.map f) ∈ Sub(D(Y)).

Definition 4.1. The four *position predicates* on a morphism f: X ⟶ Y are:

- IsInfrastructure(f) := img_f ≤ a_Y
- IsDistribution(f) := (img_f ⊓ a_Y ≠ ⊥) ∧ (img_f ⊓ a_Y^{c} ≠ ⊥)
- IsExploitation(f) := (img_f ≤ a_Y^{cc}) ∧ ¬(img_f ≤ a_Y)
- IsRefusal(f) := img_f ≤ a_Y^{c}

where ⊥ is the bottom subobject in Sub(D(Y)), and complementation and double-negation are taken in the Heyting algebra of Sub(D(Y)).

Each predicate captures a specific structural relationship between the morphism's D-image and the kernel image. Infrastructure is containment in the kernel image. Distribution is non-trivial straddling of the kernel image and its complement. Exploitation is containment in the closure-residue: inside the double-negation closure but not in the kernel image itself. Refusal is containment in the strict complement.

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5. The Partition Theorem

Theorem 5.1. Let C be an elementary topos satisfying the hypothesis bundle of Section 2. Let (D, η, ι) be a non-trivial distinction structure on C. Then for every morphism f: X ⟶ Y in C such that img_f ≠ ⊥ in Sub(D(Y)), exactly one of IsInfrastructure(f), IsDistribution(f), IsExploitation(f), IsRefusal(f) holds.

Proof. The proof consists of two parts: disjointness and exhaustiveness.

Disjointness. For each pair of position predicates, we show their conjunction is incompatible with img_f ≠ ⊥.

Infrastructure and Refusal disjoint. Suppose IsInfrastructure(f) ∧ IsRefusal(f). Then img_f ≤ a_Y ⊓ a_Y^c = ⊥, contradicting img_f ≠ ⊥.

Infrastructure and Exploitation disjoint. IsExploitation(f) requires ¬(img_f ≤ a_Y), which directly contradicts IsInfrastructure(f) := img_f ≤ a_Y.

Infrastructure and Distribution disjoint. IsInfrastructure(f) gives img_f ≤ a_Y, hence img_f ⊓ a_Y^c ≤ a_Y ⊓ a_Y^c = ⊥, contradicting the second conjunct of IsDistribution(f).

Refusal and Exploitation disjoint. Suppose IsRefusal(f) ∧ IsExploitation(f). Then img_f ≤ a_Y^c ⊓ a_Y^{cc} = ⊥ (using the Heyting identity a^c ⊓ a^{cc} = ⊥), contradicting img_f ≠ ⊥.

Refusal and Distribution disjoint. IsRefusal(f) gives img_f ≤ a_Y^c, hence img_f ⊓ a_Y ≤ a_Y^c ⊓ a_Y = ⊥, contradicting the first conjunct of IsDistribution(f).

Exploitation and Distribution disjoint. Suppose IsExploitation(f) ∧ IsDistribution(f). IsExploitation(f) gives img_f ≤ a_Y^{cc}. The first conjunct of IsDistribution(f) gives img_f ⊓ a_Y ≠ ⊥. The second conjunct gives img_f ⊓ a_Y^c ≠ ⊥. But img_f ≤ a_Y^{cc} implies img_f ⊓ a_Y^c ≤ a_Y^{cc} ⊓ a_Y^c = ⊥, contradicting the second conjunct.

Exhaustiveness. Let f satisfy img_f ≠ ⊥. We proceed by case analysis on whether img_f ≤ a_Y, and on the meet structure of img_f with a_Y and its complement.

Case 1. img_f ≤ a_Y. Then IsInfrastructure(f) holds.

Case 2. ¬(img_f ≤ a_Y) and img_f ⊓ a_Y = ⊥. In any Heyting algebra, b ⊓ a = ⊥ ↔ b ≤ a^c. Hence img_f ≤ a_Y^c, so IsRefusal(f) holds.

Case 3. ¬(img_f ≤ a_Y) and img_f ⊓ a_Y ≠ ⊥ and img_f ⊓ a_Y^c ≠ ⊥. Then IsDistribution(f) holds.

Case 4. ¬(img_f ≤ a_Y) and img_f ⊓ a_Y ≠ ⊥ and img_f ⊓ a_Y^c = ⊥. The third condition gives img_f ≤ a_Y^{cc}. Combined with ¬(img_f ≤ a_Y), this gives IsExploitation(f).

The four cases are exhaustive: they cover all combinations of the binary conditions (img_f ≤ a_Y) and (img_f ⊓ a_Y = ⊥) and (img_f ⊓ a_Y^c = ⊥). Disjointness was established above. □

Remark 5.2. The exhaustiveness of the four cases relies on the trichotomy in the Heyting algebra of subobjects: any subobject b is either contained in a, or has trivial meet with a, or has non-trivial meet with a without being contained. The fourth case (Exploitation) splits the third by checking the meet with the complement. This case structure is exhaustive in any Heyting algebra but produces non-trivial distinct classes only when the algebra is non-Boolean.

Remark 5.3. In a Boolean topos, where a^{cc} = a for all a ∈ Sub(X), the Exploitation predicate becomes (img_f ≤ a_Y) ∧ ¬(img_f ≤ a_Y), which is unsatisfiable. The partition reduces to three classes. The four-position structure is therefore specifically a feature of non-Boolean intuitionistic topoi.

Remark 5.3′ (the partition is lopsided, and the logic forces it). Infrastructure (img_f ≤ a) and Refusal (img_f ≤ a^c) are mirror images across the kernel. Exploitation (img_f ≤ a^{cc} with img_f ≰ a) has no mirror image.

The would-be fourth-cell-for-the-outside, $\mathrm{img_f} \le (a^c)^{cc}$ with $\mathrm{img_f} \not\le a^c$, is empty in every Heyting algebra, because $(a^c)^{cc} = a^c$ (triple negation collapses). The complement of any element is regular, so the outside of a distinction is always crisp; only the inside can acquire a closure-residue. Spencer-Brown's marked/unmarked asymmetry, which the distinction-structure axioms encode by fiat in η, is therefore also a consequence of the subobject logic alone: the marked side is the one that can carry a halo. (The identity $a^{ccc} = a^c$ is standard, e.g. `compl_compl_compl` in Mathlib; the remark is the one-line consequence, not stated in the literature surveyed in §7.)

Remark 5.4. The proof of Theorem 5.1 does not invoke the idempotency or coherence axioms of Definition 3.1; the partition argument uses only the action of D on morphisms, the unit η at Y, and the Heyting structure of Sub(D(Y)). The theorem therefore holds for any endofunctor-with-unit (D, η) on C satisfying the hypothesis bundle of Section 2. The full distinction-structure apparatus encodes the intended Spencer-Brown semantics — applying the distinction operation to an already-distinguished object is structurally trivial — but the partition itself is purely lattice-theoretic. The axioms are nonetheless *structurally constraining* at the level of admissible (D, η, ι): a constructive investigation (see `examples/construction-choice.md`) finds that many natural endofunctor-with-unit candidates on small toposes — the codomain-projection endofunctor on Set^⟶, constant functors at the subobject classifier, hand-engineered constant-image variants — fail the coherence condition η_{D(X)} ≫ ι.hom_X = 1_{D(X)}. A weakening of Definition 3.1 to "endofunctor with unit" would broaden the theorem's hypothesis class substantially, admitting pathological structures alongside the Spencer-Brown-faithful ones. The axioms are doing work at the level of *which structures are admissible* even though they are not load-bearing in the proof itself.

Remark 5.5. An *idempotent monad* on C is a monad (T, η, μ) with μ: T ∘ T ⟶ T a natural isomorphism. Idempotent monads on C correspond, up to equivalence, to reflective subcategories D ⊆ C: the Eilenberg-Moore category of an idempotent monad is the reflective subcategory of η-fixed objects, and conversely every reflection composed with inclusion produces an idempotent monad (Borceux 1994, Vol. 1, §4.2). Every idempotent monad satisfies the conditions of Definition 3.1 (taking ι.hom := μ; the left unit law gives coherence and μ being an iso gives idempotency). Conversely, Definition 3.1 imposes only the left unit law and idempotency, not the right unit law or associativity, so distinction structures are *a priori* more general than idempotent monads; the well-behaved cases coincide.

The connection identifies a canonical source of distinction structures: every reflective subcategory of every elementary topos provides one, and in particular sheafification for any Lawvere-Tierney topology j on C is an idempotent monad. The partition theorem can therefore be read as a statement about morphism structure induced by reflection onto any reflective subcategory of an elementary topos — a categorical interpretation that situates the apparatus inside mainstream topos theory rather than treating the (D, η, ι) signature as bespoke framework apparatus.

The corresponding non-vacuity question — *does some concrete reflective subcategory of some elementary topos produce a configuration in which all four cells of the partition are simultaneously inhabited?* — is open. Phase 1.1 of an in-progress investigation (`examples/construction-choice.md`) establishes that the Sierpinski topos Set^⟶ is too small for this: it admits exactly five Lawvere-Tierney topologies (the trivial topology, the double-negation topology, two intermediate topologies whose sheaves are trivial, and the maximal topology), of which only the double-negation case gives a non-trivial sheafification, and that sheafification lands in the Boolean part of the topos and collapses Exploitation. A non-vacuous witness requires either a richer base topos or a non-sheafification reflective subcategory. The investigation is ongoing.

A *necessary condition on the witness object* is now kernel-checked and worth recording, because it constrains where to look. Since the kernel image is a_Y := Im(η_Y), and Im(η_Y) = ⊤ whenever η_Y is an isomorphism, Refusal (img ≤ a_Y^c = ⊥) is empty at every Y where the reflection unit is invertible — that is, at every j-

sheaf, and in particular at the **terminal object**, which is a sheaf for every topology. So a four-cell witness can only occur at a Y that is *not* a j-sheaf (equivalently, an object the reflection actually moves). This generalizes the Sierpinski Candidate-E failure mode (sheafification has invertible unit on sheaves) into a topos-independent constraint. The Lean form is `kernelImage_eq_top_of_isIso_unit`; its specialization to the terminal is the reason the otherwise-natural escalation to a presheaf topos on a small poset — e.g. Set^{P^op} for P the three join-irreducibles of the subgroup lattice of $\mathbb{Z}/12$, whose terminal subobject lattice Sub(1) is the kernel-checked six-element non-Boolean Heyting algebra Div12 (`music-anchor/mazzola-bridge-note.md`; `subobjectTerminalEquivDiv12`, `kernelImage_eq_top_of_isIso_unit`) — does *not* yield a sheafification witness *at the terminal*: the witness, if one exists in such a topos, must sit at a non-terminal, non-sheaf object. That refinement of the search is the live Phase-1.2 question.

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6. Formalization

The theorem and supporting infrastructure are formalized in Lean 4 against Mathlib4 at a recent pin. The formalization comprises:

- A definition of `DistinctionStructure` as a structure type bundling D, η, and ι with the coherence condition.
- Position predicates `IsInfrastructure`, `IsDistribution`, `IsExploitation`, `IsRefusal` as defined in Section 4.
- Supporting lemmas for pairwise disjointness, each reducing to the relevant Heyting identity.
- The partition theorem `four_position_partition` proving exhaustive disjoint partition.

Kernel verification through `#print axioms` confirms that `four_position_partition` and all supporting theorems depend on exactly the standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`) with no `sorry` declarations in the transitive dependency closure. The formalization is available at the project repository (Brink 2026).

The two formalizable claims of Remarks 5.3 and 5.5 — the partition-level Boolean collapse and the idempotent-monad bridge — are mechanized in a separate companion module `FalseWork.Positions.SpencerBrown`. `boolean_partition_three_cells` lifts `exploitation_requires_nonBoolean` (the cell-level Boolean collapse) through `four_position_partition` to obtain a three-cell exhaustive disjoint partition under the local Boolean hypothesis on the subobject lattice of D(Y). `DistinctionStructure.ofIdempotentMonad` constructs a distinction structure from any Mathlib `Monad C` whose multiplication is a natural isomorphism, with the coherence condition discharged via the monad's left-unit law. Both are kernel-checked against the standard three axioms; neither closes the open construction question of Remark 5.5 in the reverse direction (whether every distinction structure of Definition 3.1 arises from an idempotent monad), which remains an open Zulip question at the time of writing.

A companion module `FalseWork.Lattice.FourPositionLattice` isolates the Heyting-algebra core of Theorem 5.1 as `lattice_four_position_partition` — for any `HeytingAlgebra H` and any `a x : H` with `x ≠ ⊥`, exactly one of the four position predicates holds at `(a, x)`. The topos-level `four_position_partition` is Layer L applied to `H = Sub(D Y)` with `a = Im(η_Y)` and `x = Im(D.map f)`. A concrete inhabitation of Layer L on a music-derived Heyting algebra lives at `FalseWork.Lattice.Examples.Div12`: the divisor lattice of 12 (equivalently the subgroup lattice of `Z/12`, equivalently the lattice of transposition-symmetric pitch-class subsets) is built as an explicit 6-element non-Boolean Heyting algebra; the witness `music_anchor_witness` establishes all four cells are inhabited at the tritone kernel by musically-meaningful subgroups (Infrastructure = tritone, Refusal = augmented triad, Exploitation = diminished 7th, Distribution = whole-tone hexachord and chromatic). Both are kernel-checked against the standard three axioms; `tritone_non_regular` is decidable and uses no axioms. The companion document `music-anchor/feasibility.md` §12 records the layered (lattice / topos / distinction structure) status; Layer T (topos realization of this lattice via `Sh(L)` or the join-irreducible presheaf topos) and

Layer D (a concrete idempotent monad delivering the kernel) are cited from general topos theory and deferred respectively.

A further companion module `FalseWork.Positions.CanonizationClosure` formalizes the framework's canonization apparatus in two related conditional theorems. `recursive_partition` is the *closure layer*: given a canonization-closure witness (data of an idempotent monad with iso multiplication) for a morphism `f`, the four-position partition theorem applies to the induced distinction structure on the ambient topos, producing a secondary four-cell classification of every morphism relative to the canonization figure. `canonization_separation` is the *generator refinement*: when the canonization data additionally certifies that the codomain `Y` of `f` is a separator (in the standard categorical sense — maps out of `Y` distinguish parallel morphisms), parallel morphisms in `C` agreeing under all `Y`-mediated comparisons are equal. Together the two theorems articulate the framework's claim that a Commitment-yes morphism organizes a structural territory (closure) within which it acts as a measuring instrument (generator). Both are kernel-checked against the standard three axioms; the substantive open question is the closure-side predicate identifying *which* idempotent monads count as canonization closures of a given morphism (three candidates surveyed in the companion document `closure-canonization.md`, selection deferred pending future work), while the generator-side separator condition is mathematically standard and concretely formalizable. A first kernel-checked concrete `CanonizationGenerator` witness (identity monad on `Discrete PUnit`) lives at `FalseWork.Positions.Examples.DiscretePUnit.canonizationGenerator`, establishing type-level inhabitation; richer instantiations remain open. A conditional cross-layer alignment definition `CanonizationGenerator.ofCommitmentYes` records the framework's architectural claim that a Commitment-yes morphism with a separator codomain admits a canonization-generator witness; the definition is kernel-checked but its proof does not yet use the commitment hypothesis because `IsCommitmentYes` is currently a schema-level placeholder. See `closure-canonization.md` §§8a–8b for the full discussion.

The Heyting algebra structure on Sub(X) for elementary topoi — required by the partition theorem but not present in Mathlib at the time of this work — was submitted upstream as Pull Request #39618 to mathlib4 (Brink 2026, PR) and closed unmerged on 2026-05-28 under the project's new-contributor AI-use policy; the instance remains kernel-checked in this development. The PR implements the Mac Lane–Moerdijk equalizer-residual construction (Mac Lane and Moerdijk 1992, IV.6 Proposition 2): implication is defined as `Subobject.mk (equalizer.ι (χ (P ⊓ Q).arrow) (χ P.arrow))`, and the Galois connection $R \le P \Rightarrow Q \leftrightarrow R \sqcap P \le Q$ is established through six private bridging lemmas. The hypothesis bundle required by the PR matches that of Theorem 5.1.

The development was undertaken with AI collaboration using Cursor with Anthropic Claude as the underlying model, in accordance with the project's documented validation architecture (Brink 2026, Paper 2). The kernel verification is independent of AI assistance: Lean's kernel mechanically checks the proof against the type theory regardless of how the proof terms were constructed.

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7. Motivation, Related Work, and Scope

7.1 Motivation

The partition theorem emerged from the development of the FalseWork project (Brink 2026, Paper 1), a cross-domain structural analysis framework that proposes practitioners across creative and intellectual domains distribute into a small number of structural positions relative to generative operations that produce irreducible features. The four positions of the theorem are intended to formalize this structural typology in the categorical setting where it admits precise statement.

The framework operates with the intuition that an asymmetry in a domain forces a distinction operation, which when iterated produces a kernel whose failure to close on itself generates an irreducible structural gap.

The four positions represent the structurally distinct ways practitioners can relate to this gap: by operating in the territory the kernel reaches (Infrastructure), by distributing across the boundary (Distribution), by working in the asymmetric closure-residue (Exploitation), or by operating in the strict complement (Refusal).

The theorem establishes that whenever this structural intuition can be formalized as a non-trivial distinction structure on an elementary topos, the four positions arise as a necessary and exhaustive partition. The framework's broader claim — that specific domains across music, cinema, painting, literature, software, and physics instantiate this structural condition — is an empirical claim about each domain that the theorem does not itself establish.

7.2 Related Work

Spencer-Brown. *Laws of Form* (Spencer-Brown 1969) introduces the calling axiom and the crossing axiom as foundational moves for a calculus of distinctions. Monadic idempotency $D \ggg D \cong D$ in categorical settings is a natural translation of the calling axiom. Prior categorical work on Spencer-Brown's calculus includes Kauffman (1987) and the broader cybernetic tradition. Kauffman's treatments are oriented toward recursive forms, knot-theoretic encodings, and the algebra of self-reference rather than the functor-composition idempotency rendering used here; the framework's specific rendering — the calling axiom as the idempotency component of an endofunctor-with-unit structure, applied to elementary topoi — follows the framework's development (Brink 2026, Paper 1, Paper 3) and does not appear in the standard topos theory or Spencer-Brown literature surveyed by the author. A companion document `spencer-brown-anchor.md` develops the structural correspondence in detail: Definition 3.1 lifts the calling axiom into category theory; the partition theorem's four cells read as four Spencer-Brown registers (under the mark, straddling the mark, in the failure-of-crossing residue, crossed out); and Exploitation is identified as the cell that exists precisely because Spencer-Brown's crossing axiom (the Boolean condition $\neg\neg p = p$) fails in non-Boolean Heyting algebras.

Mac Lane and Moerdijk. The Heyting algebra structure on subobjects of an elementary topos (Mac Lane and Moerdijk 1992, IV.6, IV.8) is the standard reference for the construction the partition theorem depends on. The PR #39618 formalization follows IV.6 Proposition 2 directly.

Mathematical music theory. The framework's music application instantiates the partition theorem on a music topos drawn from existing infrastructure rather than constructing one independently. The relevant prior apparatus includes Mazzola's topos-theoretic framework for music (Mazzola 2002), the categorical reading of Lewin's transformational theory (Lewin 1987; Mazzola and Andreatta 2006), the Poly-Klumpenhouwer Network apparatus (Popoff, Andreatta, and Ehresmann 2018), Tymoczko's voice-leading geometry and groupoid formalization of musical space (Tymoczko 2011, 2026), the lattice-and-topology representation of musical relations via formal concept analysis and Q-analysis (Freund, Andreatta, and Giavitto 2017), and the application of mathematical-morphology operators to concept lattices (Atif, Bloch, Distel, and Hudelot 2013; for an overview situating these within the SMIR project, see Andreatta 2018). The structural-typological reading of the four cells in terms of practitioner-positions is the framework's contribution; the underlying categorical infrastructure on which the music application instantiates is drawn from this body of prior work and used instrumentally rather than re-derived. A standalone music-anchor paper is in preparation (see `music-anchor/feasibility.md`).

Reflective subcategories and idempotent monads. As recorded in Remark 5.5, the distinction structures of Definition 3.1 are closely related to idempotent monads on C, and idempotent monads on a category correspond to reflective subcategories of it (Borceux 1994, Vol. 1, §4.2; Adámek and Rosický 1994). Sheafification for a Lawvere-Tierney topology on a topos is the canonical example. The framing of the partition theorem as a statement about reflection-induced morphism partitions — i.e., a structural statement about how morphisms of an elementary topos distribute relative to the kernel image of the reflection unit, organized by Heyting position in the parent topos's subobject lattice — does not appear in the standard literature on reflective subcategories

or sheafifications surveyed by the author. Standard treatments focus on the algebraic properties of reflections (closure under limits, the localization equivalence, the Eilenberg-Moore correspondence) rather than on the closure-residue / non-Boolean structure of subobject lattices in the parent topos.

Categorical logic and topos theory. Standard references include Johnstone (2002a, 2002b), Awodey (2010), Lawvere (1969). The non-Boolean character of subobject lattices in elementary topoi is well-established; the framing as a structural partition with the specific Heyting conditions of Section 4 has not been located in the standard literature.

7.3 Scope

The theorem establishes a structural partition in elementary topoi with non-trivial distinction structure. Several questions are deliberately not addressed.

The theorem does not establish that any specific domain instantiates the structural condition. Whether music, cinema, painting, literature, software, physics, or other domains have the categorical structure the theorem requires is empirical work that the framework pursues separately (Brink 2026, Paper 3 on music; Paper 5 on the Pythagorean comma specifically). Two anchor scoping documents accompany this preprint: `music-anchor/feasibility.md` records the music-side layered (lattice / topos / distinction structure) status, with the lattice layer kernel-checked in Lean (`FalseWork.Lattice.Examples.Div12.music_anchor_witness`); `physics-anchor/feasibility.md` scopes the physics-side target via Bohrification / topos quantum mechanics (Heunen-Landsman-Spitters 2009, Döring-Isham 2007), with no Lean layer kernel-checked in this round and a Wolfram Route-B exploration of small finite physics-interpretable lattices on record. Both documents are exploratory by design and the framework does not claim either anchor closes the empirical question of domain instantiation.

The theorem does not validate classifications of specific works or practitioners as occupying specific positions. The framework's broader claims about Bach as Infrastructure, Coltrane as Exploitation, Schoenberg as Refusal, and similar classifications across domains are supported by specialist correspondence (Tymoczko, Cutting, Suhr) and by structural analysis of specific works, but they require domain-level construction of the relevant categorical settings to be made formal.

The theorem does not address the framework's *Commitment gate* — a binary predicate operating orthogonally within each position to characterize structural completion. The gate is documented as schema-level architecture in the framework's papers (Brink 2026, `comma-formal-structure-note`) and has been shown through formalization testing to not admit theorem-grade unification of its four position-specific extension operators.

The theorem does not claim novelty as topos theory in any strong sense. The mathematical building blocks are standard. The framing as a structural partition with these specific Heyting conditions tied to the distinction-structure apparatus does not appear in the standard topos theory literature the author has surveyed; specialist literature search has not been undertaken and prior work in adjacent registers may exist that has not been located.

Postscript: companion results (September 2026, v1.1)

Three developments postdating the May draft sharpen the theorem's standing; all are kernel-checked in the same repository and none changes anything above.

1. **The non-degeneracy criterion.** The four cells are simultaneously inhabited, at the lattice level, if and only if the kernel element is *ordinary* in Citkin's sense (neither regular nor dense) — so the partition's non-degeneracy has an exact algebraic characterization, and any Heyting algebra carrying a non-degenerate partition has at least six elements and order-embeds Z_6. Kernel-checked (`isOrdinary_iff_allFourCells`,

`Positions/OrdinaryKernel.lean`); full development with prior-art adjudication in the companion preprint `preprints/ordinary-elements-z6/`.

2. **A concrete inhabited instance from threshold logic.** In Opens $\mathbb{R}$, a single threshold unit's decision region is a *regular* element (the partition around it is degenerate — a lone perceptron is classical), while a two-layer composed region is *ordinary*, with all four cells inhabited and the double-negation remainder computed exactly (`Examples/PerceptronRegular.lean`). This supplies what §7.3′s scope note asked for: a naturally occurring, non-constructed instance where the four-fold provably opens. Companion note: `preprints/perceptron-bridge/`.
3. **Observer-relative refinement.** The partition's kernel-dependence is developed into two kernel-checked invariants — the *aperture* (which coarse-grainings keep the four-fold open) and the *co-aperture* (how much confusion each coarse-graining manufactures), independent of each other in both directions — in `preprints/aperture/`.

How to cite

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The Lean formalization is part of the citable object: `lean/FalseWorkPapers/Positions/` at github.com/thefalsework/papers.

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